

```

1 //adder is been updated to do add on both Left and Right
2 module Adder (
3     //Left side of the adder
4     input wire [23:0] dataout_L,
5     input wire [23:0] shift_outL,
6     input wire      add_sub_L, // 0: add, 1: subtract
7     output wire [23:0] adder_outL,
8
9     //Right side of the adder
10    input wire [23:0] dataout_R,
11    input wire [23:0] shift_outR,
12    input wire      add_sub_R, // 0: add, 1: subtract
13    output wire [23:0] adder_outR
14);
15    assign adder_outL = add_sub_L ? (shift_outL - dataout_L) : (shift_outL + dataout_L);
16    assign adder_outR = add_sub_R ? (shift_outR - dataout_R) : (shift_outR + dataout_R);
17 endmodule
18
19
20 `timescale 1ns / 1ps
21
22 module Adder_tb;
23
24     // Inputs
25     reg signed [23:0] dataout_L, shift_outL;
26     reg signed [23:0] dataout_R, shift_outR;
27     reg add_sub_L, add_sub_R;
28
29     // Outputs
30     wire signed [23:0] adder_outL, adder_outR;
31
32     // Instantiate DUT
33     Adder UUT (
34         .dataout_L(dataout_L),
35         .shift_outL(shift_outL),
36         .add_sub_L(add_sub_L),
37         .adder_outL(adder_outL),
38
39         .dataout_R(dataout_R),
40         .shift_outR(shift_outR),
41         .add_sub_R(add_sub_R),
42         .adder_outR(adder_outR)
43     );
44
45     initial begin
46         $display("Starting Adder Testbench...");
47         $monitor("Time=%0t | L: %0d %s %0d = %0d | R: %0d %s %0d = %0d",
48             $time,
49             shift_outL, add_sub_L ? "-" : "+", dataout_L, adder_outL,
50             shift_outR, add_sub_R ? "-" : "+", dataout_R, adder_outR);
51
52         // Test Case 1: Add both sides
53         shift_outL = 24'sd100;
54         dataout_L  = 24'sd25;
55         add_sub_L  = 0;
56
57         shift_outR = 24'sd200;
58         dataout_R  = 24'sd50;
59         add_sub_R  = 0;
60         #10;
61
62         // Test Case 2: Subtract both sides
63         add_sub_L = 1;
64         add_sub_R = 1;
65         #10;
66
67         // Test Case 3: Negative operands
68         shift_outL = -24'sd100;
69         dataout_L  = 24'sd50;

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70     add_sub_L   = 0;
71
72     shift_outR  = -24'sd80;
73     dataout_R   = -24'sd20;
74     add_sub_R   = 0;
75     #10;
76
77     // Test Case 4: Subtract with negatives
78     add_sub_L = 1;
79     add_sub_R = 1;
80     #10;
81
82     // Test Case 5: Max and min values
83     shift_outL  = 24'sh7FFFFFF; // Max positive
84     dataout_L   = 24'sh800000; // Min negative
85     add_sub_L   = 0;
86
87     shift_outR  = 24'sh800000;
88     dataout_R   = 24'sh7FFFFFF;
89     add_sub_R   = 1;
90     #10;
91
92     $display("Testbench finished.");
93     #10 $finish;
94 end
95
96 endmodule
97
```